

Unit - III Knowledge and Reasoning

Teaching Hours: 06

Marks: 12



Knowledge-Based agent

- An Agent who have the capability of maintaining an internal state of knowledge, reason over that knowledge, update their knowledge after observations and take actions.
- These agents can represent the world with some formal representation and act intelligently.
- Knowledge-based agents are composed of two main parts:
 - **Knowledge-base and**
 - **Inference system.**

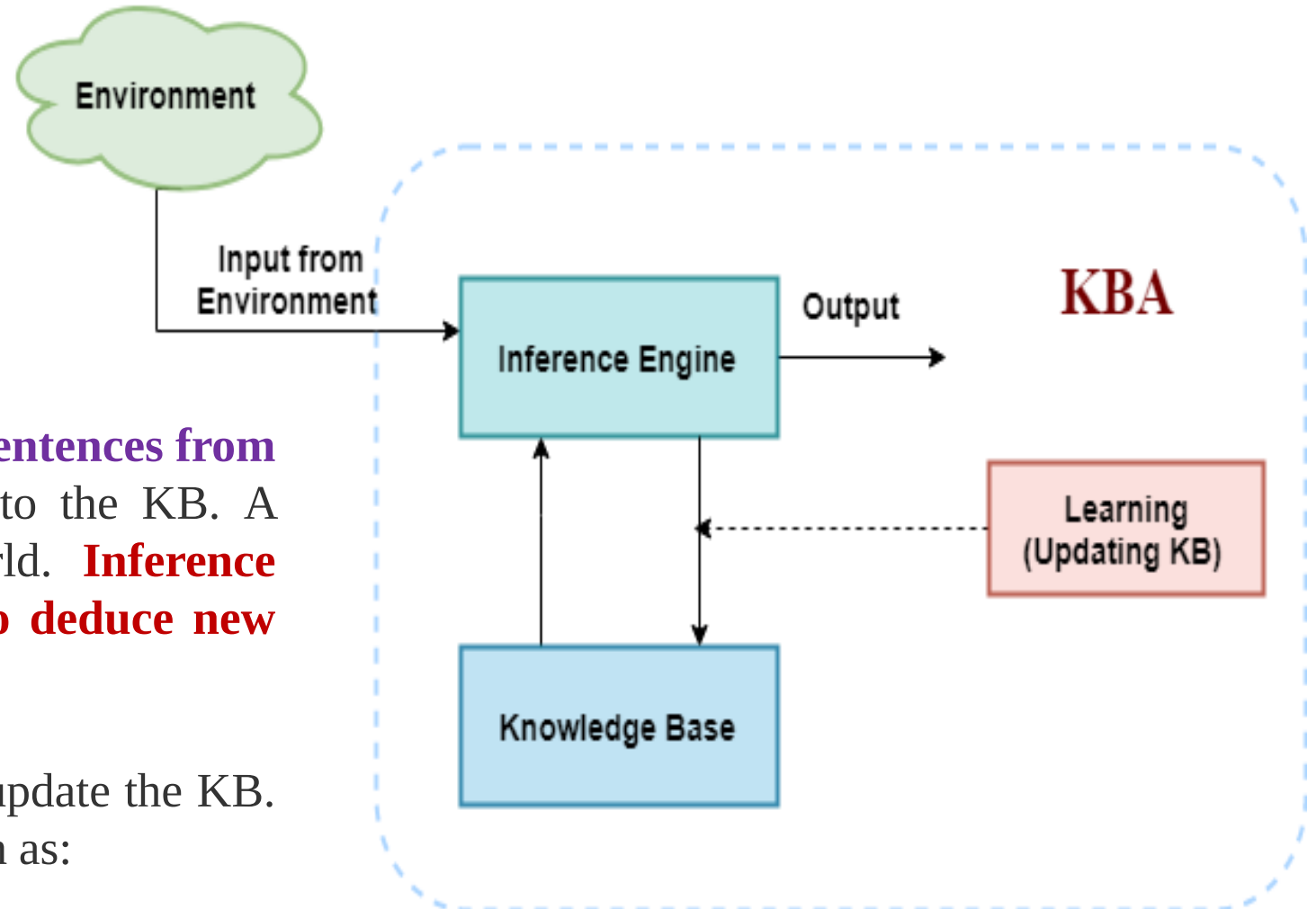
The architecture of knowledge-based agent(KBA):

Knowledge base: It is a central component of a **KBA**, it is also known as KB. It is a **collection of sentences**. These sentences are expressed in a language which is called a knowledge representation language. The Knowledge-base of KBA **stores fact about the world**.

Inference system: It means **deriving new sentences from old**. It allows us to add a new sentence to the KB. A sentence is a proposition about the world. **Inference system applies logical rules to the KB to deduce new information**.

It generates new facts so that an agent can update the KB. It works mainly in two rules which are given as:

- Forward chaining
- Backward chaining



Operations Performed by KBA

- **TELL:** This operation tells the knowledge base what it perceives from the environment.
- **ASK:** This operation asks the knowledge base what action it should perform.
- **Perform:** It performs the selected action.

Various levels of knowledge-based agent

- **1. Knowledge level**

- Knowledge level is the **first level of knowledge-based agent**, in this level, we need to specify what the agent knows, and what the agent goals are.
- Suppose, an automated taxi agent needs to go from a station A to station B, and he knows the way from A to B, so this comes at the knowledge level.

- **2. Logical level:**

- At the logical level, an encoding of knowledge into logical sentences occurs.
- At the logical level we can expect to the automated taxi agent to reach to the destination B.

- **3. Implementation level:**

- At the implementation level agent perform actions as per logical and knowledge level.
- At this level, an automated taxi agent actually implement his knowledge and logic so that he can reach to the destination.

What is knowledge representation?

- Humans are best at understanding, reasoning, and interpreting knowledge.
- Human knows things, which is knowledge and as per their knowledge they perform various actions in the real world.
- **How machines do all these things? It comes under knowledge representation and reasoning.**
- Agent can utilize this knowledge to solve the complex real world problems.
- **Knowledge Representation** is not just storing data into some database, but it also enables an agent to learn from that knowledge.

What to Represent:

- **Object:** All the facts about objects in our world domain. E.g., Guitars contains strings, trumpets are brass instruments.
- **Events:** Events are the actions which occur in our world.
- **Performance:** It describe behavior which involves knowledge about how to do things.
- **Meta-knowledge:** It is knowledge about what we know.
- **Facts:** Facts are the truths about the real world and what we represent.
- **Knowledge-Base:** The central component of the KBA is the knowledge-base. It is represented as KB. The Knowledgebase is a group of the Sentences.

Types of knowledge

- 1. Declarative Knowledge
- 2. Procedural Knowledge
- 3. Meta-knowledge
- 4. Heuristic knowledge
- 5. Structural knowledge

Types of knowledge Conti....

- **1. Declarative Knowledge:**

- Declarative knowledge is to **know about something**.
- It includes **concepts, facts, and objects**.
- It is also called descriptive knowledge and expressed in declarative sentences.
- It is simpler than procedural language.

- **2. Procedural Knowledge**

- It is also known as **imperative knowledge**.
- Procedural knowledge is a type of knowledge which is responsible for knowing **how to do something**.
- It can be directly **applied to any task**.
- It includes **rules, strategies, procedures, agendas**, etc.
- Procedural knowledge depends on the task on which it can be applied.

Types of knowledge Conti....

- **3. Meta-knowledge:**

Knowledge about the other types of knowledge is called Meta-knowledge.

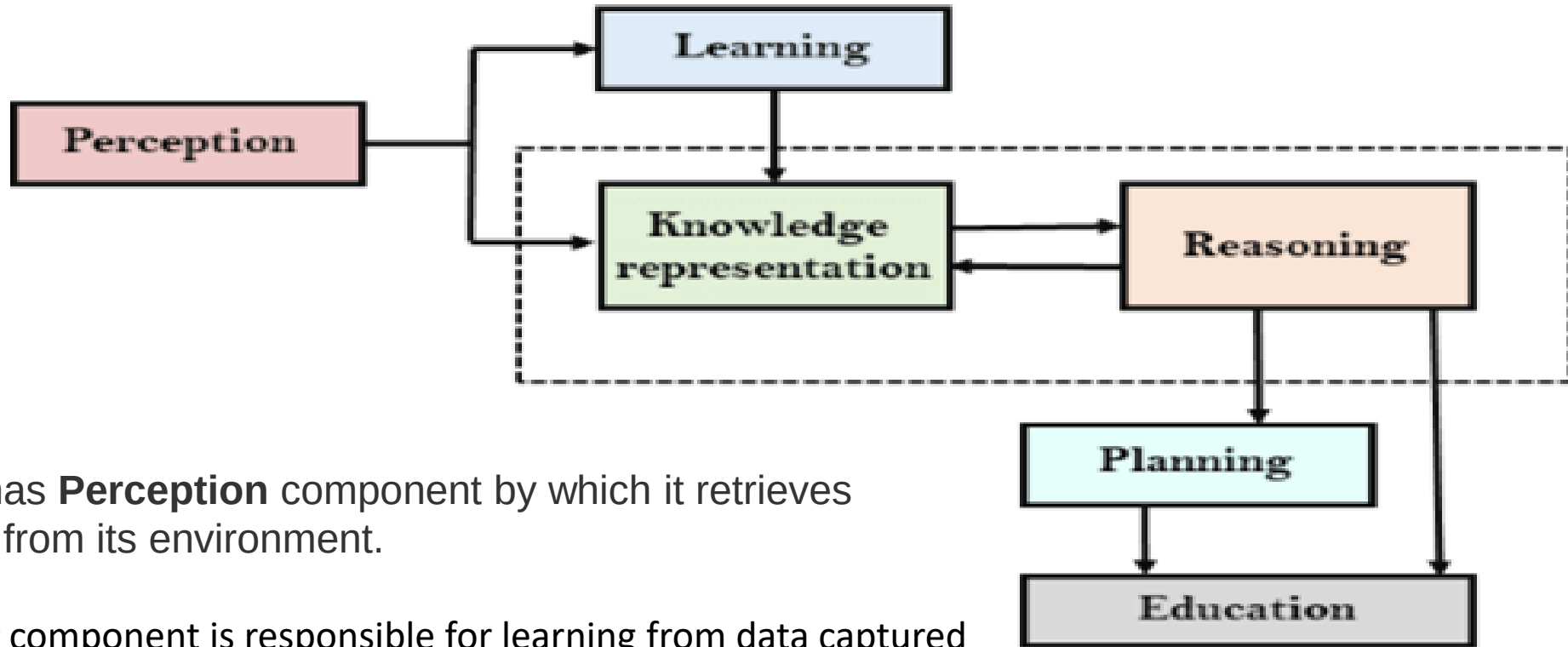
- **4. Heuristic knowledge:**

- Heuristic knowledge is representing **knowledge of some experts in a field or subject**.
- Heuristic knowledge is **rules of thumb based on previous experiences**, awareness of approaches, and which are good to work but not guaranteed.

- **5. Structural knowledge:**

- Structural knowledge is basic knowledge to problem-solving.
- It describes **relationships between various concepts** such as **kind of, part of, and grouping of something**.
- It describes the **relationship** that **exists between concepts or objects**.

AI knowledge cycle:



- AI system has **Perception** component by which it retrieves information from its environment.
- **The learning** component is responsible for learning from data captured by Perception component
- **Knowledge Representation and Reasoning** components are involved in showing the intelligence in machine-like humans.

Approaches to designing a knowledge-based agent

- There are mainly two approaches to build a knowledge

1. Declarative Knowledge:

- We can create a knowledge-based agent by providing facts and rules in the knowledge base instead of programming every action directly.
- designer builds the agent by telling it what it needs to know. The knowledge is stored as facts and rules in the Knowledge Base (KB). The agent then uses reasoning (inference) to decide what action to perform.

2. Procedural Knowledge

- It is also known as **imperative knowledge**.
- Procedural knowledge is a type of knowledge which is responsible for knowing **how to do something**.
- It can be directly **applied to any task**.
- It includes **rules, strategies, procedures, agendas**, etc.
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NOTE: Today's Modern AI system combine both approaches.

Approaches to knowledge representation:

1. Simple relational knowledge:
2. Inheritable knowledge:
3. Inferential knowledge:
4. Procedural knowledge:

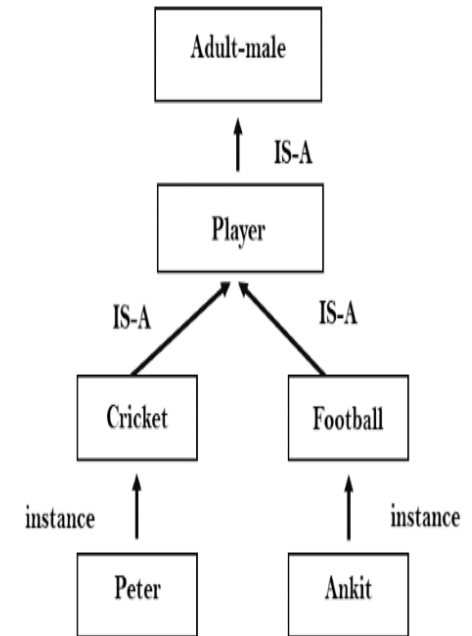
1. Simple relational knowledge:

- It is the simplest way of storing facts which uses the relational method, and each fact about a set of the object is set out systematically in columns.
- This approach is famous in database systems where the relationship between different entities is represented.

Player	Weight	Age
Player1	65	23
Player2	58	18
Player3	75	24

2. Inheritable knowledge:

- In this, all data must be stored into a hierarchy of classes.
- All classes should be arranged in a generalized form or a hierarchal manner.
- This approach contains inheritable knowledge which shows a relation between **instance and class**, and it is called instance relation.
- Every individual frame can represent the collection of attributes and its value.
- **Objects and values** are represented in **Boxed nodes**.
- We use **Arrows which point from objects to their values**.



3. Inferential knowledge:

- Inferential knowledge approach represents knowledge in the form of formal logics.
- This approach can be used to derive more facts.
- It guaranteed correctness.

Example: Let's suppose there are two statements:

- Mohan is a man
- All men are mortal

Then it can represent as;

man(Mohan)

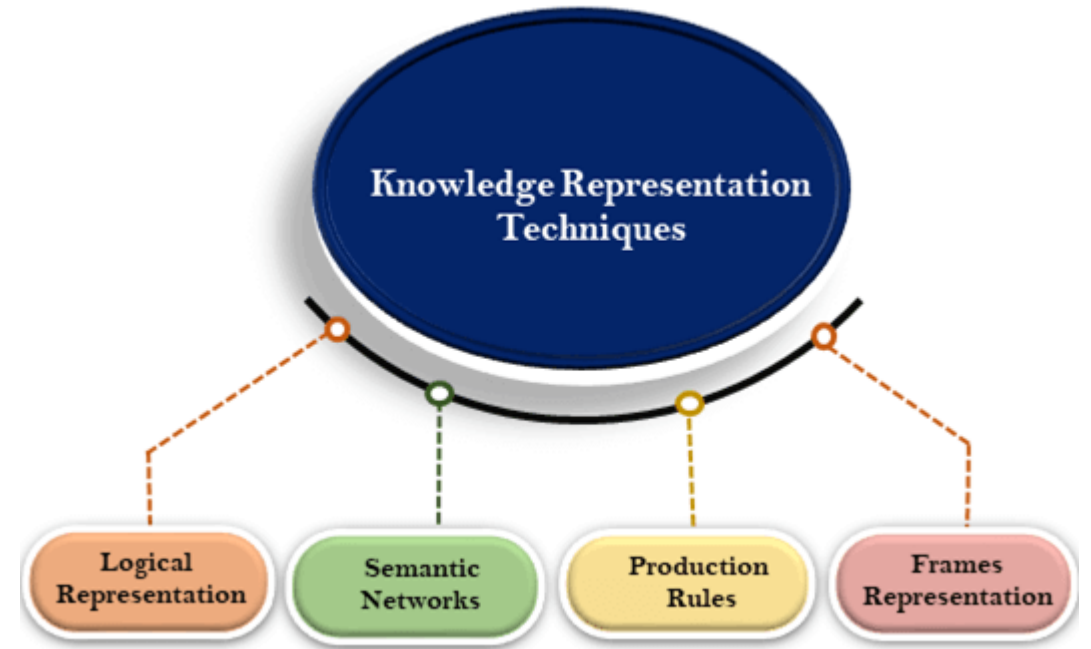
$\forall x = \text{man}(x) \text{ -----} \rightarrow \text{mortal}(x)$

4. Procedural knowledge:

- It **uses small programs and codes** which describes how to do specific things, and how to proceed.
- **If-Then rule** is used here.
- Various coding languages such as **LISP language** and **Prolog language** used.
- Heuristic or domain-specific knowledge easily represented here.
- There is no need to represent all cases.

Techniques of knowledge representation

- Logical Representation
- Semantic Network Representation
- Frame Representation
- Production Rules



Logical Representation

- Logical representation means drawing a conclusion based on various conditions.
- This representation lays down some important communication rules. It consists of precisely defined syntax and semantics.
- Logical representation can be categorized into mainly two logics:
 - Propositional Logics
 - Predicate logics(First Order Logic)

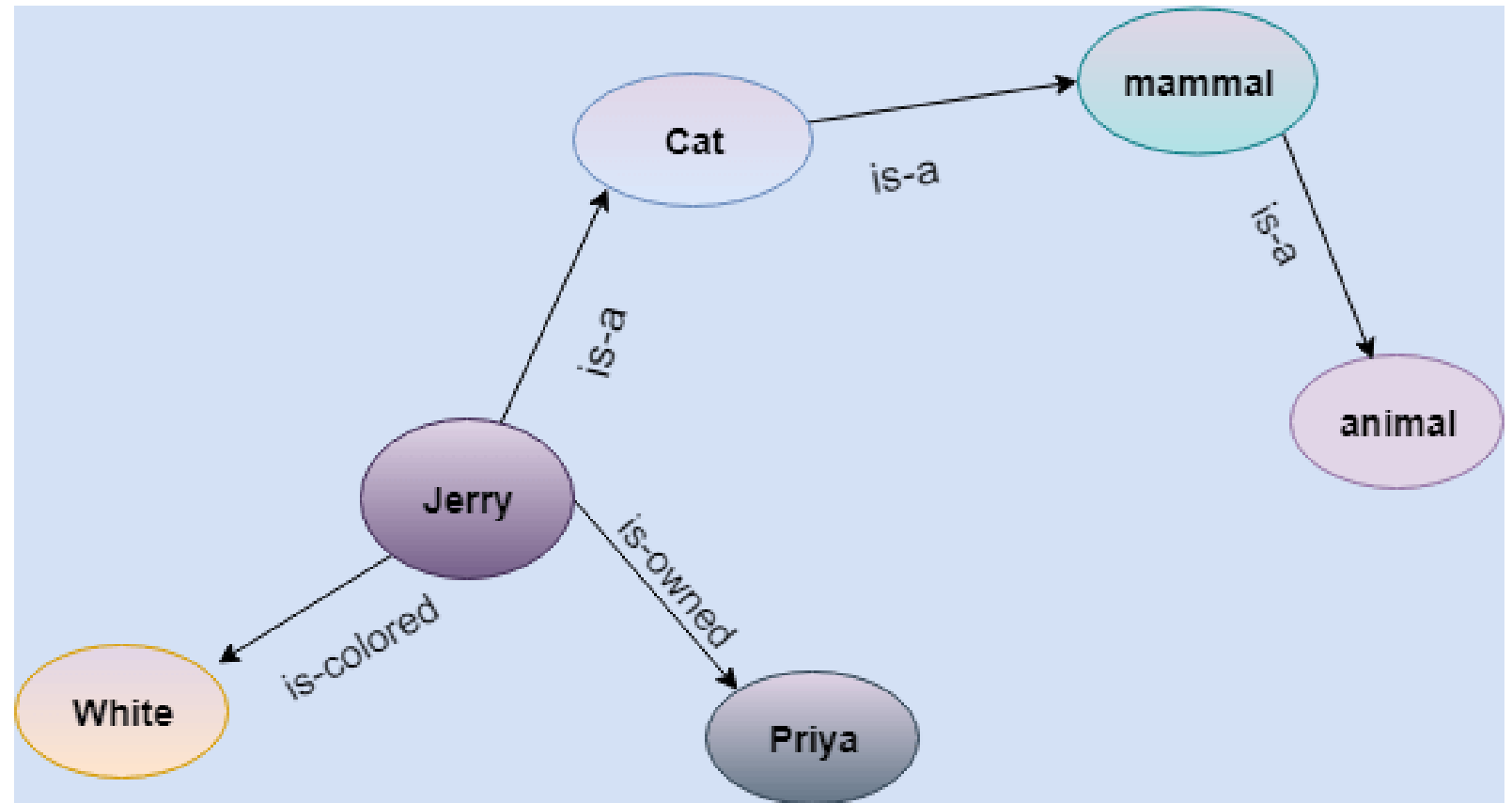
2. Semantic Network Representation

- It is an alternative of predicate logic for knowledge representation.
- Knowledge can represent in the form of graphical networks.
- This network consists of nodes representing objects and arcs which describe the relationship between those objects.
- Semantic networks can categorize the object in different forms and can also link those objects.
- This representation consist of mainly two types of relations:
 - IS-A relation (Inheritance)
 - Kind-of-relation

Example: Following are some statements which we need to represent in the form of nodes and arcs.

Statements:

1. Jerry is a cat.
2. Jerry is a mammal
3. Jerry is owned by Priya.
4. Jerry is White colored.
5. All Mammals are animal.



3. Frame Representation

- A frame is a **record like structure** which consists of a collection of attributes and its values to describe an entity in the world.
- **Frames** are the **AI data structure** which divides knowledge into substructures by representing stereotypes situations.
- It consists of a **collection of slots and slot values**. These slots may be of any type and sizes. Slots have names and values which are called facets.
- **Facets:** The various aspects of a slot is known as **Facets**.
- Facets are features of frames which enable us to put constraints on the frames. Example: IF-NEEDED facts are called when data of any particular slot is needed.
- A frame may consist of any number of slots, and a slot may include any number of facets and facets may have any number of values. A frame is also known as **slot-filter knowledge representation** in artificial intelligence.

Frame for a book

Slots	Filters
Title	Artificial Intelligence
Genre	Computer Science
Author	Peter Norvig
Edition	Third Edition
Year	1996
Page	1152

4. Production Rules

- Production rules system consist of (**condition, action**) pairs which mean, "If condition then action". It has mainly three parts:
 - The set of production rules
 - Working Memory
 - The recognize-act-cycle
- In production rules agent checks for the condition and if the condition exists then production rule fires and corresponding action is carried out.
- The condition part of the rule determines which rule may be applied to a problem.
- And the action part carries out the associated problem-solving steps. This complete process is called a **recognize-act cycle**.
- The working memory contains the description of the current state of problems-solving and rule can write knowledge to the working memory. This knowledge match and may fire other rules.
- If there is a new situation (state) generates, then multiple production rules will be fired together, this is called conflict set. In this situation, the agent needs to select a rule from these sets, and it is called a conflict resolution.

Example:

- IF (at bus stop AND bus arrives) THEN action (get into the bus)
- IF (on the bus AND paid AND empty seat) THEN action (sit down).
- IF (on bus AND unpaid) THEN action (pay charges).
- IF (bus arrives at destination) THEN action (get down from the bus).

Propositional logic

- It is a technique of knowledge representation in logical and mathematical form.
- It is the simplest form of logic where all the statements are made by propositions.
- A proposition is a declarative statement which is either true or false.
- Examples:
 - a) It is Sunday.
 - b) The Sun rises from West (False proposition)
 - c) $3+3=7$ (False proposition)
 - d) 5 is a prime number.

Syntax of propositional logic

The syntax of propositional logic defines the allowable sentences for the knowledge representation. There are two types of Propositions:

- Atomic Propositions
- Compound propositions

•**Atomic Proposition:** Atomic propositions are the simple propositions. It consists of a single proposition symbol. These are the sentences which must be either true or false.

Example:

- a) $2+2$ is 4, it is an atomic proposition as it is a **true** fact.
- b) "The Sun is cold" is also a proposition as it is a **false** fact.

•**Compound proposition:** Compound propositions are constructed by combining simpler or atomic propositions, using parenthesis and logical connectives.

Example:

- a) "It is raining today, and street is wet."
- b) "Ankit is a doctor, and his clinic is in Mumbai."

Propositional Logic Connectives

Connective symbols	Word	Technical term	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only if	Biconditional	$A \Leftrightarrow B$
$\neg$ or $\sim$	Not	Negation	$\neg A$ or $\neg B$

Propositional Logic Connectives

- **Conjunction:** Rohan is intelligent and hardworking.
P= Rohan is intelligent,
Q= Rohan is hardworking. $\rightarrow P \wedge Q$.
- **Disjunction:** "Ritika is a doctor or Engineer",
Here P= Ritika is Doctor. Q= Ritika is Doctor, so we can write it as $P \vee Q$.
- **Implication:** If it is raining, then the street is wet.
Let P= It is raining, and Q= Street is wet, so it is represented as $P \rightarrow Q$
- **Biconditional:** If I am breathing, then I am alive
P= I am breathing, Q= I am alive, it can be represented as $P \Leftrightarrow Q$.

Tautology:

- A tautology is a compound proposition that is always true, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to true.

if any compound proposition is always T $\Rightarrow$ Tautology

Tautology:

- Example $P \vee \neg P$
Always True tautology:

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

(always true)

Tautology:

- Example $(P \wedge Q) \rightarrow P$ (Yes) ← Tautology.
when ever it is F then implies is T.

P ✓	Q ✓	$P \wedge Q$	$(P \wedge Q) \rightarrow P$? F
F.	F.	F.	T
F.	T.	F.	T
T.	F.	F.	T
T.	T.	T.	T

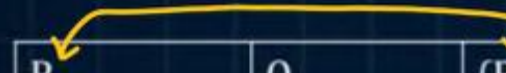
Tautology:

- Example $P \rightarrow (Q \rightarrow P) \Rightarrow$ Tautology. ✓

P	Q	$Q \rightarrow P$?	$P \rightarrow (Q \rightarrow P)$ F
T	T.	T	T
T	F	T	T
F	T.	F	T
F	F	T	T

Tautology:

- Example $P \vee (P \wedge Q) \leftrightarrow P$.



P	Q	$(P \wedge Q)$	$P \vee (P \wedge Q)$	P	$P \vee (P \wedge Q) \leftrightarrow P$
F	F	F	F	F	T
F	T	F	F	F	T
T	F	F	T	T	T
T	T	T	T	T	T

Tautology:

- Example $\neg (P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$

P	Q	$\neg P$	$\neg Q$	$(\neg P \vee \neg Q)$	$\neg (P \wedge Q)$	$\neg (P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$
F	F	T	T	T	T	T
F	T	T	F	T	T	T
T	F	F	T	T	T	T
T	T	F	F	F	F	T

Contradiction:

- Contradiction in Propositional Logic
- A contradiction is a compound proposition that is always false, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to false.

→ The proposition which are always False ✓

Contradiction:

- A contradiction is the opposite of a tautology. It reflects a logical impossibility- something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.

$$(P \wedge (P \rightarrow Q) \wedge \neg Q)$$

$\neg Q$
F
T
F
T

P Q
T T
T F
F F
F T

$P \rightarrow Q$
T
F
T
T

$P \wedge (P \rightarrow Q)$
T
F
F
F

$$(P \wedge (P \rightarrow Q) \wedge \neg Q)$$

F
F
F
F

→ Contradiction

$$(\neg (P \rightarrow Q) \wedge P \wedge Q)$$

P	Q	$(P \rightarrow Q)$	$\neg (P \rightarrow Q) \wedge P \wedge Q$
F	F	T	F
F	T	T	F
T	F	F	F
T	T	T	F

Contingency in Propositional Logic:

- A contingency is a compound proposition that is neither always true (a tautology) nor always false (a contradiction). Its truth value depends on the truth values of its atomic propositions, meaning it can be either true or false based on different assignments of truth values.
- Explanation: Contingent propositions are the most common type of propositions in logic because their truth depends on the specific conditions. Unlike tautologies and contradictions, which have fixed truth values, contingencies vary based on the situation.

Contradiction in Propositional Logic:

- A contradiction is the opposite of a tautology. It reflects a logical impossibility—something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.

Contingency in Propositional Logic:

- A contradiction is the opposite of a tautology. It reflects a logical impossibility—something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.
- **Identifying a Contingency Truth Table:** A proposition is a contingency if its truth table contains both true and false outcomes. This means the truth value depends on specific cases. Logical Status: Contingencies are neither universal truths (like tautologies) nor universal falsehoods (like contradictions).

Contingency in Propositional Logic:

- Example 1: $(P \wedge Q)$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Examples:

Contradiction

- Contradiction in Propositional Logic
- A contradiction is a compound proposition that is always false, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to false.

Contradiction

- A contradiction is the opposite of a tautology. It reflects a logical impossibility—something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.

#Q. $(\neg(P \rightarrow Q) \wedge P \wedge Q)$

Contradiction.

P	Q	$P \rightarrow Q$	$\neg(P \rightarrow Q)$	$P \wedge Q \wedge \neg(P \rightarrow Q)$
F	F	T	F	F
F	T	T	F	F
T	F	F	T	F
T	T	T	F	F

#Q. What is $((P \rightarrow Q) \wedge (Q \rightarrow P))$

$P \leftrightarrow Q$ equivalence Bi conditional

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
F	F	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	T	T	T

First-Order Logic

- In propositional logic, we can only represent the facts, which are either true or false.
- PL is not sufficient to represent the complex sentences or natural language statements.
 - "Some humans are intelligent", or
 - "Sachin likes cricket."
- FOL is sufficiently expressive to represent the natural language statements in a concise way.
- FOL assumes the following things in the world:
 - **Objects:** A, B, people, numbers, colors, wars, theories, squares, pits
 - **Relations:** It can be **unary relation such as:** red, round, is adjacent, **or n-any relation such as:** the sister of, brother of, has color, comes between
 - **Function:** Father of, best friend, third inning of, end of,
- FOL has two main parts:
 - **Syntax**
 - **Semantics**

Syntax of First-Order logic

- The syntax of FOL determines which collection of symbols is a logical expression in first-order logic.
- We write statements in short-hand notation in FOL.
- Basic Elements of FOL

Constant	1, 2, A, John, Mumbai, cat,....
Variables	x, y, z, a, b,....
Predicates	Brother, Father, >,....
Function	sqrt, LeftLegOf,
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$
Equality	$=$
Quantifier	$\forall$, $\exists$

Atomic sentences

- Atomic sentences are the most basic sentences of FOL.
- These sentences are formed from a predicate symbol followed by a parenthesis with a sequence of terms.
- FOL statements can be divided into two parts:
 - **Subject:** Subject is the main part of the statement.
 - **Predicate:** A predicate can be defined as a relation, which binds two atoms together in a statement.
- We can represent atomic sentences as Predicate (term1, term2, ..., term n).
- **Example:**

Ravi and Ajay are brothers: $\Rightarrow$ Brothers(Ravi, Ajay).



Chinky is a cat: $\Rightarrow$ cat (Chinky).

Universal Quantifier

- Universal quantifier is a **symbol of logical representation**, which specifies that the statement within its range is true for everything or every instance of a particular thing.
- The Universal quantifier is **represented by a symbol $\forall$** , which resembles an inverted A.
- If x is a variable, then $\forall x$ is read as:
 - **For all x**
 - **For each x**
 - **For every x .**
- Example:
 - **All man drink coffee.** $\forall x \text{ man}(x) \rightarrow \text{drink}(x, \text{coffee}).$
It will be read as: **There are all x where x is a man who drink coffee.**

Existential Quantifier

- Existential quantifiers are the type of quantifiers, which express that the **statement within its scope is true for at least one instance of something**.
- It is denoted by the **logical operator $\exists$** , which resembles as inverted E.
- When it is used with a predicate variable then it is called as an existential quantifier.
- In Existential quantifier we always use **AND or Conjunction symbol ($\wedge$)**.
- If x is a variable, then existential quantifier will be $\exists x$ or $\exists(x)$. And it will be read as:
 - There exists a 'x.'
 - For some 'x.'
 - For at least one 'x.'
- Example:

Some boys are intelligent. $\exists x: \text{boys}(x) \wedge \text{intelligent}(x)$

It will be read as: **There are some x where x is a boy who is intelligent.**

Examples:

Eg:

- a) Ravi likes all kind of food.
- b) Apple and chicken are food.
- c) Anything anyone eats and is not killed is food
- d) Ajay eats peanuts and still alive.
- e) Rita eats that Ajay eats.

$$\forall x : \text{food}(x) \longrightarrow \text{likes}(\text{Ravi}, x)$$

$$\text{i) food}(\text{apple}) \quad \text{ii) food}(\text{chicken})$$

$$\forall x \forall y : \text{eats}(x, y) \wedge \neg \text{killed}(x)$$

$$\text{eats}(\text{ajay}, \text{peanuts}) \wedge \text{alive}(\text{ajay})$$

$$\forall c : \text{eats}(\text{Ajay}, c) \rightarrow \text{eats}(\text{Rita}, c)$$

Examples:

1. All birds fly.

$\forall x \text{ bird}(x) \rightarrow \text{fly}(x).$

2. Every man respects his parent.

$\forall x \text{ man}(x) \rightarrow \text{respects}(x, \text{parent}).$

3. Some boys play cricket.

$\exists x \text{ boys}(x) \rightarrow \text{play}(x, \text{cricket}).$

4. Not all students like both Mathematics and Science.

$\neg \forall (x) [\text{student}(x) \rightarrow \text{like}(x, \text{Mathematics}) \wedge \text{like}(x, \text{Science})].$

5. Only one student failed in Mathematics.

$\exists (x) [\text{student}(x) \rightarrow \text{failed}(x, \text{Mathematics}) \wedge \forall (y) [\neg(x=y) \wedge \text{student}(y) \rightarrow \neg \text{failed}(y, \text{Mathematics})].$

Examples:

#Q. Which of the following formulas is a formalization of the sentence: "Every connected and circuit-free graph is a tree"
Where $\text{cir}(x)$ means "x has circuit".

$\neg \text{Cir}(x)$ ✓

- A** $\forall x [(\text{conn}(x) \wedge \neg \text{cir}(x)) \rightarrow \text{tree}(x)]$
- B** $\exists \neg x [\text{conn}(x) \wedge \neg \text{cir}(x)] \wedge \neg \text{tree}(x)$
- C** $\forall x [(\text{conn}(x) \wedge \neg \text{cir}(x)) \wedge \text{tree}(x)]$
- D** $\forall x [(\text{conn}(x) \wedge \text{cir}(x)) \rightarrow \text{tree}(x)]$

Examples:

#Q. We use the dictionary:

M domain of men

s Sheenu

N(x) x is nice

L(x, y) x loves y

Which of the following formulas corresponds to the sentence:

There is a nice man who loves Sheenu.

(A) $\exists x \in M [N(x) \wedge L(x, s)]$

(B) $\exists x \in M [N(x) \rightarrow L(x, s)]$

(C) $\forall x \in M [N(x) \wedge L(x, s)]$

(D) $\forall x \in M [N(x) \rightarrow L(x, s)]$

Examples:

#Q. Assume the domain is $\mathbb{R}$. Which of the following statements are true?

- I. $\forall x \exists y (xy = 1)$
- II. $\exists x \forall y (x^2 \geq y^2)$
- III. $\exists x \exists y (x^2 \geq y^2)$

false

$$\forall x \exists y P(x, y) \Rightarrow \exists y \forall x P(x, y)$$

at least one y for all x

$$\bullet \forall x \exists y (xy = 1)$$

for all x there exist at least one y ($xy = 1$) Yes

A Only (ii).

$$\exists x \forall y (x^2 \geq y)$$

at least one x satisfy condition for all y. false

B Only (iii).

$$\exists x \exists y (x^2 \geq y) \Rightarrow \text{to make it true only one pair of } x, y$$

(2, 1)

C Only (i) and (ii).

D Only (i) and (iii).

Free and Bound Variables

Free Variable:

A variable is said to be a free variable in a formula if it occurs outside the scope of the quantifier.

Example:

$\forall x \exists (y)[P(x, y, z)]$, where z is a free variable.

Bound Variable:

A variable is said to be a bound variable in a formula if it occurs within the scope of the quantifier.

Example:

$\forall x [A(x) B(y)]$, here x and y are the bound variables.

Argument

- The statement that A set of premises $(P_1, P_2, \dots, P_n)$ yields another proposition
- Q is called an Argument.
- Q is called the conclusion of the Argument.
- So, this is trying to say that when $P_1, P_2, P_3, \dots, P_n$ are true then Q is also true....

Valid Argument $\Rightarrow$ All Premise are true then Conclusion must be true

Argument

Example-1:

- P: Every student of Information Technology studies Data Structures.
- P: Data structures necessarily contain the study of arguments.
- C: Therefore, every student of Information Technology studies arguments.

If both Premise are true then 'C' true??

Yes

Argument

Example-2:

- Every parent is a mature person.
- Children should listen to mature people.
- Therefore, every child should listen to their parents.

Valid.

Rule of Inference

- **Modus ponens (MP):**
- Also known as affirming the antecedent, this rule allows you to infer the consequent if the antecedent is true.
- Form: $P \Rightarrow Q$
- P (Premise P is true)
- Conclusion: Q (Therefore, Q is true)

Premise: $P \rightarrow Q$

Premise: P

Conclusion: Q must be true

$P \rightarrow Q$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Rule of Inference

Example:

- Premise 1 : If it rains, the streets will be wet
- Premise 2 : It is raining
- Conclusion : The streets will be wet.

$P \rightarrow Q$	P	Q
T	T	T
T	T	F
F	F	T
T	F	F

Rule of Inference

- Modus Tollens (MT):
- Also known as denying the consequent, this rule states that if the consequent is false, the antecedent must also be false.
- Form: $P \Rightarrow Q$ (If P, then Q)
- $\neg Q$ (Not Q)
- Conclusion: $\neg P$ (Not P)

$P_1: A \rightarrow B$

$P_2: \neg B$

Conc: $\neg A$

Rule of Inference

- Example: $Q \rightarrow P$
- Premise 1 : If it rains, the streets will be wet.
- Premise 2 : The streets are not wet. $\neg P$

Conclusion :

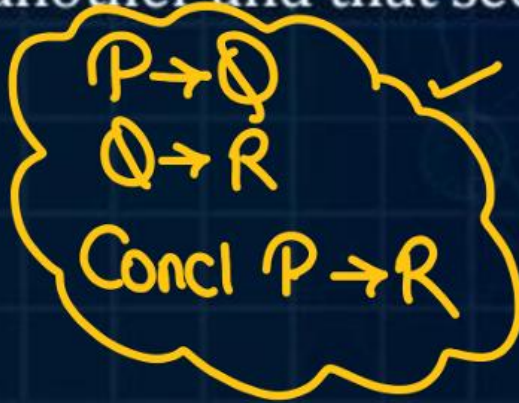
It is not raining. $\neg Q$

- P : Street are wet
- Q : it is Raining

P	Q	$Q \rightarrow P$	$\neg P$	$\neg Q$
F	F	T	T	T
F	T	F	T	F
T	F	T	F	T
T	T	T	F	F

Rule of Inference *

- Hypothetical Syllogism (HS):
- This rule allows you to infer a direct relationship between two statements if one statement leads to another and that second statement leads to a third.
- Form : $P \Rightarrow Q$
 $Q \Rightarrow R$
- Conclusion: $P \Rightarrow R$



Rule of Inference

Example:

- Premise 1: If I study, I will pass the exam. $P \rightarrow Q$
- Premise 2: If I pass the exam I will graduate. $Q \rightarrow R$
- Conclusion: If I study, I will graduate. $P \rightarrow R$

Rule of Inference

- 4. Disjunctive Syllogism (DS):
 - This rule is used when one of two possibilities is true, and one of them is false, allowing you to infer the other must be true.
 - Form: $P \vee Q$ (Either P or Q)
 - $\neg P$ (Not P)
 - Conclusion: Q (Therefore, Q)
- $P \vee Q$
 - $\neg P$
 - then Conclusion Q.

Rule of Inference

- Example:
- Premise 1: Either I will have coffee or tea. $A \vee B$
- Premise 2: I will not have coffee. $\neg B$
- Conclusion: I will have tea. $\rightarrow A$

Valid

- A: I will have Tea
- B: I will have Coffee.

A	B	P ₁ : $A \vee B$	P ₂ : $\neg B$	Concl: A
T	T	T	F	T
T	F	T	T	T
F	T	F	F	F
F	F	F	T	F

Rule of Inference

- Addition:
 - This rule allows you to add a disjunction (logical OR) with any statement.
 - Form: P
 - Conclusion: $P \vee Q$
- Premise P .
- Conclusion $P \vee Q$

Rule of Inference

- Example:
- Premise: The sun is shining.
- Conclusion: The sun is shining, or it is raining.

A : The sun is shining

B : It is Raining

Valid.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Diagram illustrating the truth table for the disjunction $A \vee B$. The table shows the truth values of A (The sun is shining) and B (It is Raining) and the resulting truth value of the disjunction $A \vee B$. The disjunction is true in all cases except when both A and B are false. Red circles and arrows highlight the first two rows where the disjunction is true.

Rule of Inference

- **Simplification**: This rule allows you to infer one part of a conjunction.
- Form: $P \wedge Q$ (Both P and Q are true)
- Conclusion: P

Premise: $P \wedge Q$
Concl: P

✓

Rule of Inference

- Example:
- Premise: I have a pen and a notebook.
- Conclusion: I have a pen.

Valid.

Rule of Inference

- **Conjunction:**
- This rule allows you to combine two true statements into a single conjunction.
- Form: **P**
- **Q**
- Conclusion: $P \wedge Q$ ✓

Rule of Inference

- Example:
- Premise 1: **I have a pen.**
- Premise 2: **I have a notebook.**
- Conclusion: **I have a pen and a notebook.** ✓

Rule of Inference

- Resolution:
- A rule used in disjunctive reasoning to infer new conclusions from disjunctions.
- Form: $P \vee Q$
- $\neg P \vee R$
- Conclusion: $Q \vee R$

• $P \vee Q$
• $\neg P \vee R$
Valid.

P	Q	R	$P \vee Q$	$\neg P \vee R$	$Q \vee R$
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	1

Rule of Inference

- Example:
- Premise 1: I will go for a walk or stay home. $P \vee Q$
- Premise 2: I will not go for a walk or I will read a book. $\neg P \vee R$
- Conclusion: I will stay home or read a book. $Q \vee R$

P: I will go for walk

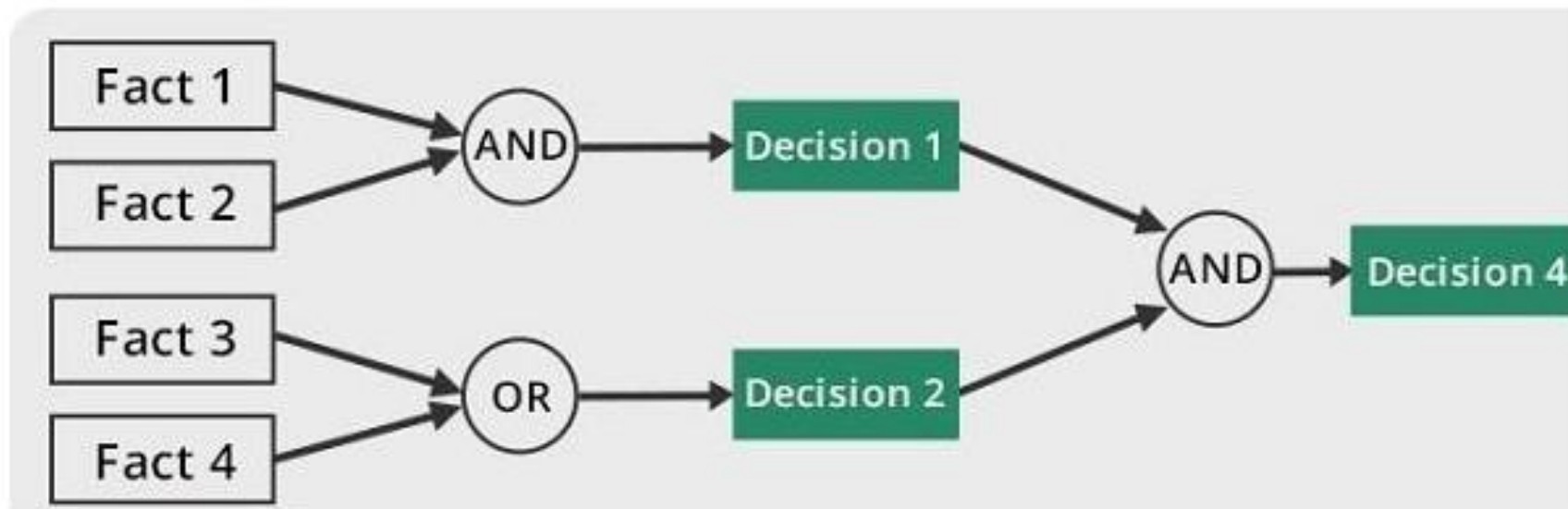
Q: I will stay home

R: I will Read book

-
- Forward Chaining
 - Backward Chaining

Forward Chaining

- Forward chaining is a method of reasoning in artificial intelligence in which inference **rules are applied to existing data** to extract additional data until an endpoint (goal) is achieved.



Forward Chaining Steps

- In the first step, the system is given one or more than one constraints.
- Then the rules are searched in the knowledge base for each constraint. The rules that fulfil the condition are selected(i.e., IF part).
- Now each rule is able to produce new conditions from the conclusion of the invoked one. As a result, THEN part is again included in the existing one.
- The added conditions are processed again by repeating step 2. The process will end if there is no new conditions exist.

Properties of forward chaining

- The process uses a **down-up approach (bottom to top)**.
- It starts from an initial state and uses facts to make a conclusion.
- This approach is data-driven.
- It's employed in expert systems and production rule system.

Examples of forward chaining

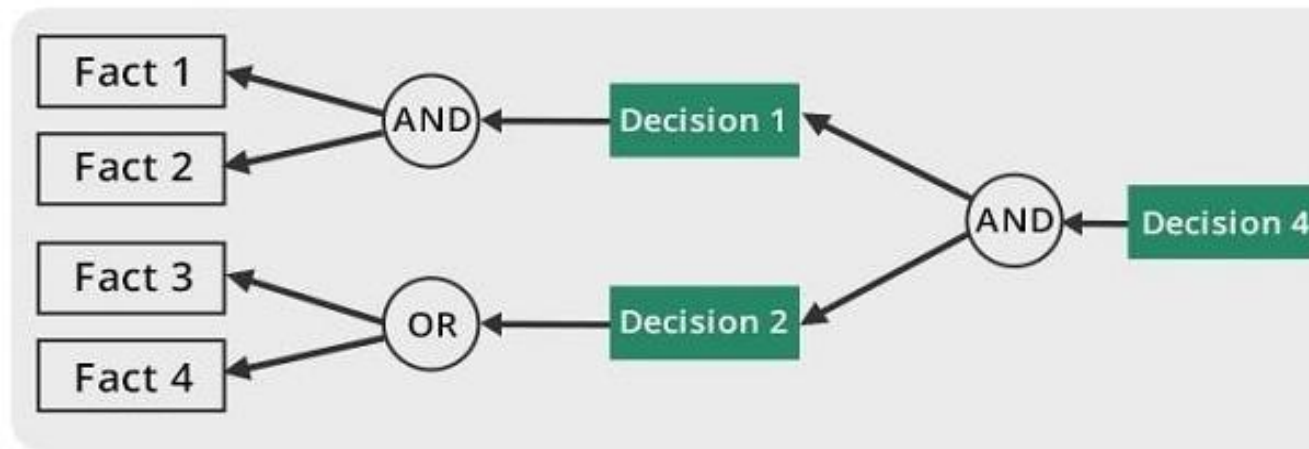
- A simple example of forward chaining can be explained in the following sequence.
- A
- $A \rightarrow B$
- B
- A is the starting point. $A \rightarrow B$ represents a fact. This fact is used to achieve a decision B.
- A practical example will go as follows;
- Tom is running (A)
- If a person is running, he will sweat ($A \rightarrow B$)
- Therefore, Tom is sweating. (B)

Examples of forward chaining

- Let's consider a medical diagnosis system where the goal is to determine if a patient has the flu. The system starts with known facts:
 - **Fact 1:** The patient has a fever.
 - **Fact 2:** The patient has a sore throat.
 - **Rule:** If the patient has a fever and sore throat, they might have the flu.
 - The system applies the rule, leading to the conclusion that the patient might have the flu.
-
- In forward chaining, facts are often represented in **First-Order Logic (FOL)** to apply rules systematically. For example:
 - **Fact:** Fever(John)
 - **Rule:** $\text{Fever}(x) \wedge \text{SoreThroat}(x) \rightarrow \text{Flu}(x)$
The system identifies **John** as the subject and applies the rule to infer that John might have the flu.

Backward Chaining

- Backward chaining is a concept in artificial intelligence that involves **backtracking from the endpoint or goal to steps that led to the endpoint.**
- This type of chaining starts from the goal and moves backward to comprehend the steps that were taken to attain this goal.



Backward Chaining Steps

- Firstly, the goal state and the rules are selected where the goal state reside in the THEN part as the conclusion.
- From the IF part of the selected rule the sub goals are made to be satisfied for the goal state to be true.
- Set initial conditions important to satisfy all the sub goals.
- Verify whether the provided initial state matches with the established states. If it fulfils the condition then the goal is the solution otherwise other goal state is selected.

Properties of backward chaining

- The process uses an up-down approach (top to bottom).
- It's a goal-driven method of reasoning.
- The endpoint (goal) is subdivided into sub-goals to prove the truth of facts.
- A backward chaining algorithm is employed in inference engines, game theories, and complex database systems.
- The [modus ponens inference rule](#) is used as the basis for the backward chaining process. This rule states that if both the conditional statement ($p \rightarrow q$) and the antecedent (p) are true, then we can infer the subsequent (q).

Example of backward chaining

- The information provided in the previous example (forward chaining) can be used to provide a simple explanation of backward chaining. Backward chaining can be explained in the following sequence.
- B
- $A \rightarrow B$
- A
- B is the goal or endpoint, that is used as the starting point for backward tracking. A is the initial state. $A \rightarrow B$ is a fact that must be asserted to arrive at the endpoint B.
- A practical example of backward chaining will go as follows:
- Tom is sweating (B).
- If a person is running, he will sweat ($A \rightarrow B$).
- Tom is running (A).

Example of backward chaining

- **Goal:** Does the patient have the flu?
- **Rule:** If the patient has a fever and sore throat, they might have the flu.
- **Sub-goals:**
 - Verify if the patient has a fever.
 - Verify if the patient has a sore throat.

Difference between Forward and Backward Chaining

Aspect	Forward Chaining	Backward Chaining
Approach	Data-Driven: Starts from facts and applies rules to reach conclusions.	Goal-Driven: Starts with a goal and works backward to verify if facts support it.
Search Strategy	Breadth-First Search: Explores multiple rules at the same level.	Depth-First Search: Focuses deeply on one path before trying others.
Direction	Bottom-Up: Moves from facts to conclusions.	Top-Down: Begins with the goal and works towards the facts.
Efficiency	May explore irrelevant rules, potentially reducing efficiency.	Risk of getting stuck in infinite loops if not managed properly.
Memory Usage	Can require more memory as it processes multiple rules at once.	More memory-efficient as it focuses on specific goals or sub-goals.
Complexity	Easier to implement for systems with many rules and data.	Can be more complex due to recursive searches for supporting facts.
Performance	Performs better when all relevant data is known upfront.	Works well when only specific information or goals are of interest.
Examples of Use Cases	Troubleshooting, diagnostics, and prediction systems.	Query systems, expert systems, and decision-making models.

Reasoning in Artificial Intelligence

What do you mean by the term “Reasoning”?

- Reasoning is an act of **deriving a conclusion from certain premises** using a given methodology.
- Reasoning is a **process of thinking; logically arguing; drawing inference.**
- **When a system is required to do something, that it has not been explicitly told how to do, it must reason.** It must figure out what it needs to know from what it already knows.
- Many types of reasoning have been identified and recognized, but many questions regarding their logical and computational properties remained under controversy.
- Popular methods of Reasoning include Abduction, Induction, deductive etc. All of them are intimately related to problems of belief revision and theory development, knowledge assimilation, discovery and learning.

- Example: Fact-1 : Robins are birds, Fact-2 : All birds have wings.

Then we can ask: DO ROBINS HAVE WINGS?

- Reasoning is an act of deriving a conclusion from certain premises using a given methodology.

- Any knowledge system must reason, if it is required to do something which has not being told explicitly.
- For reasoning system must find out, what it needs to know from what it already knows.

Hence to answer the above question- some reasoning must go.

Human reasoning capabilities

- Broadly it is being divided into three areas:
 - **Mathematical Reasoning**- *axioms, definitions, theorems, proofs*
 - **Logical Reasoning**- *deductive, inductive, abductive*
 - **Non-logical Reasoning**- *linguistic, language*
- Above three mentioned are in every human being, but the ability level depends on education, genetics and environment.
- Intelligent Quotient (IQ)= mathematical + logical reasoning, whereas Emotional Quotient mostly depends on non-logical reasoning capabilities.
- Logical Reasoning is our major concern in AI.

Reasoning in AI

- Remember the ways of KR in AI. various ways to reason on this knowledge using different logical schemes.
- **Reasoning:**
 - It is the mental process of deriving logical conclusion and making predictions from available knowledge, facts, and beliefs.

OR

- **"Reasoning is a way to infer facts from existing data."** It is a general process of thinking rationally, to find valid conclusions.

Types of Reasoning

- In AI, reasoning can be divided into the
- following categories:
 - ✓ Deductive reasoning
 - ✓ Inductive reasoning
 - ✓ Adductive reasoning
 - ✓ Common Sense Reasoning
 - ✓ Monotonic Reasoning
 - ✓ Non-monotonic Reasoning

1. Deductive reasoning

- Deductive reasoning is **deducing new information from logically related known information**.
- It is the form of valid reasoning, which means the argument's conclusion must be true when the premises are true.
- DR is a type of propositional logic in AI, and it requires various rules and facts.
- It is sometimes referred to as **top-down reasoning**, and contradictory to inductive reasoning.
- In DR, the truth of the premises guarantees the truth of the conclusion.

1. Deductive reasoning

- Deductive reasoning **mostly starts from the general premises to the specific conclusion**, which can be explained as below example.
- Example:
 - Premise-1: All the human eats vegetables.
 - Premise-2: Abebe is human.
 - Conclusion: Abebe eats veggies.

Deductive reasoning(Cont..)

A formal English statement of this rule is : If X is true and if X being true implies Y is true then Y is true. $(X \wedge (X \rightarrow Y) \rightarrow Y)$

Example: All cats are felines. Bosty is a cat. I can deduce that **Bosty is a feline**

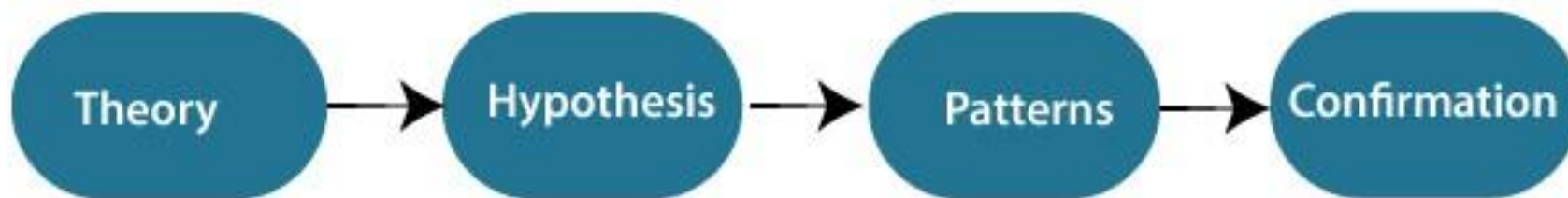
Example 2:

Premise-1: All the human eats veggies

Premise-2: Suresh is human.

Conclusion: Suresh eats veggies.

- The general process of deductive reasoning is given below:



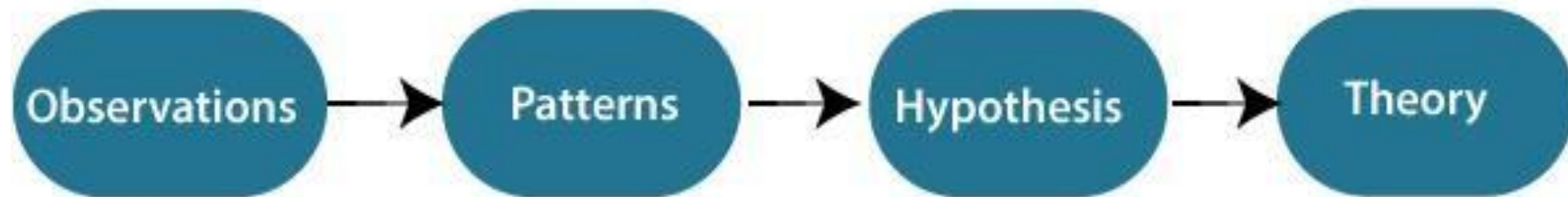
2. Inductive Reasoning

- IR is a form of reasoning to arrive at a conclusion using limited sets of facts by the process of generalization.
- It starts with the series of specific facts or data and reaches to a general statement or conclusion.
- IR is a type of propositional logic, which is also known as cause-effect reasoning or bottom-up reasoning.
- In IR, use historical data or various premises to generate a generic rule, for which premises support the conclusion.
- In IR, premises provide probable supports to the conclusion, so the truth of premises does not guarantee the truth of the conclusion.

...cont.

- **Example:**

- **Premise:** All of the pigeons we have seen in the zoo are white.
- **Conclusion:** Therefore, we can expect all the pigeons to be white.



3. Adductive reasoning

- **Adductive reasoning** is a form of logical reasoning which starts with **single or multiple observations** then **seeks to find the most likely explanation or conclusion** for the observation.
- AR is an **extension of deductive reasoning**, but in adductive reasoning, the **premises do not guarantee the conclusion.**
- **Example:**
 - **Implication:** Cricket ground is wet if it is raining
 - **Axiom:** Cricket ground is wet.
 - **Conclusion:** It is raining.

Abduction Reasoning

- Example:
- Implication:**If I read Smoking causes lung cancer and Frank died of lung cancer , I may infer that Frank was a smoker.** Again, this may or may not be true. Using statistics and probability theory , abduction may yield the most probable inference among many possible inferences.

Abduction Reasoning

1) All successful , entrepreneurial industrialists are rich persons.

2) John is a rich person.

- If this was only information available, a plausible inference would be that **John was a successful , entrepreneurial industrialist**. This conclusion could also be false since there are other roads to riches such as inheritance , the lottery...If we had a table of the income distribution of wealthy persons along with their personal histories, we could refine our abduction inference with the probability of the inference being true .
- Another Example:
- Implication: Cricket ground is wet if it is raining
- Axiom: Cricket ground is wet.
- Conclusion It is raining.

4. Common Sense Reasoning

- Common sense reasoning is **an informal form of reasoning, which can be gained through experiences.**
- CSR simulates the human ability to make presumptions about events which occurs on every day.
- It relies on good judgment rather than exact logic and operates on **heuristic knowledge** and **heuristic rules.**
- **Example:**
 - **One person can be at one place at a time.**
 - **If I put my hand in a fire, then it will burn.**
 - The above two statements are the examples of common sense reasoning which a human mind can easily understand and assume.

5. Monotonic Reasoning

- In monotonic reasoning, **once the conclusion is taken**, then it will remain the same even if we add some other information to existing information in our knowledge base.
 - In MR, **adding knowledge does not decrease the set of prepositions that can be derived.**
 - To solve monotonic problems, we can derive the **valid conclusion from the available facts only**, and it will not be affected by new facts.
 - Monotonic reasoning is **not useful for the real-time systems**, as in real time, facts get changed, so we cannot use monotonic reasoning.
 - Monotonic reasoning is used in conventional reasoning systems, and a logic-based system is monotonic.
 - Any theorem proving is an example of monotonic reasoning.

...cont.

- **Example:**
 - **Earth revolves around the Sun.**
 - It is a true fact, and it cannot be changed even if we add another sentence in knowledge base like, "The moon revolves around the earth" Or "Earth is not round," etc.
- **Advantages of Monotonic Reasoning:**
 - each old proof will always remain valid.
 - If we deduce some facts from available facts, then it will remain valid for always.
- **Disadvantages of Monotonic Reasoning:**
 - Cannot represent the real world scenarios using Monotonic reasoning.
 - Hypothesis knowledge cannot be expressed with monotonic reasoning, which means facts should be true.
 - Since we can only derive conclusions from the old proofs, so new knowledge from the real world cannot be added.

6. Non-monotonic Reasoning

- In Non-monotonic reasoning, **some conclusions may be invalidated** if we add some more information to our knowledge base.
 - Logic will be said as non-monotonic if some conclusions can be invalidated by adding more knowledge into our knowledge base.
 - Non-monotonic reasoning deals with incomplete and uncertain models. **"Human perceptions for various things in daily life, "is a general example of non-monotonic reasoning.**
- **Example:** Let suppose the KBS contains the following knowledge:
 - **Birds can fly**
 - **Penguins cannot fly**
 - **Pitty is a bird**
- So from the above sentences, we can conclude that **Pitty can fly**.
 - However, if we add one another sentence into knowledge base **"Pitty is a penguin"**, which concludes **"Pitty cannot fly"**, so it invalidates the above conclusion.

...cont.

- **Advantages of Non-monotonic reasoning:**
 - For real-world systems such as Robot navigation, we can use non-monotonic reasoning.
 - In Non-monotonic reasoning, we can choose probabilistic facts or can make assumptions.
- **Disadvantages of Non-monotonic Reasoning:**
 - In non-monotonic reasoning, the old facts may be invalidated by adding new sentences.
 - It cannot be used for theorem **proving**.

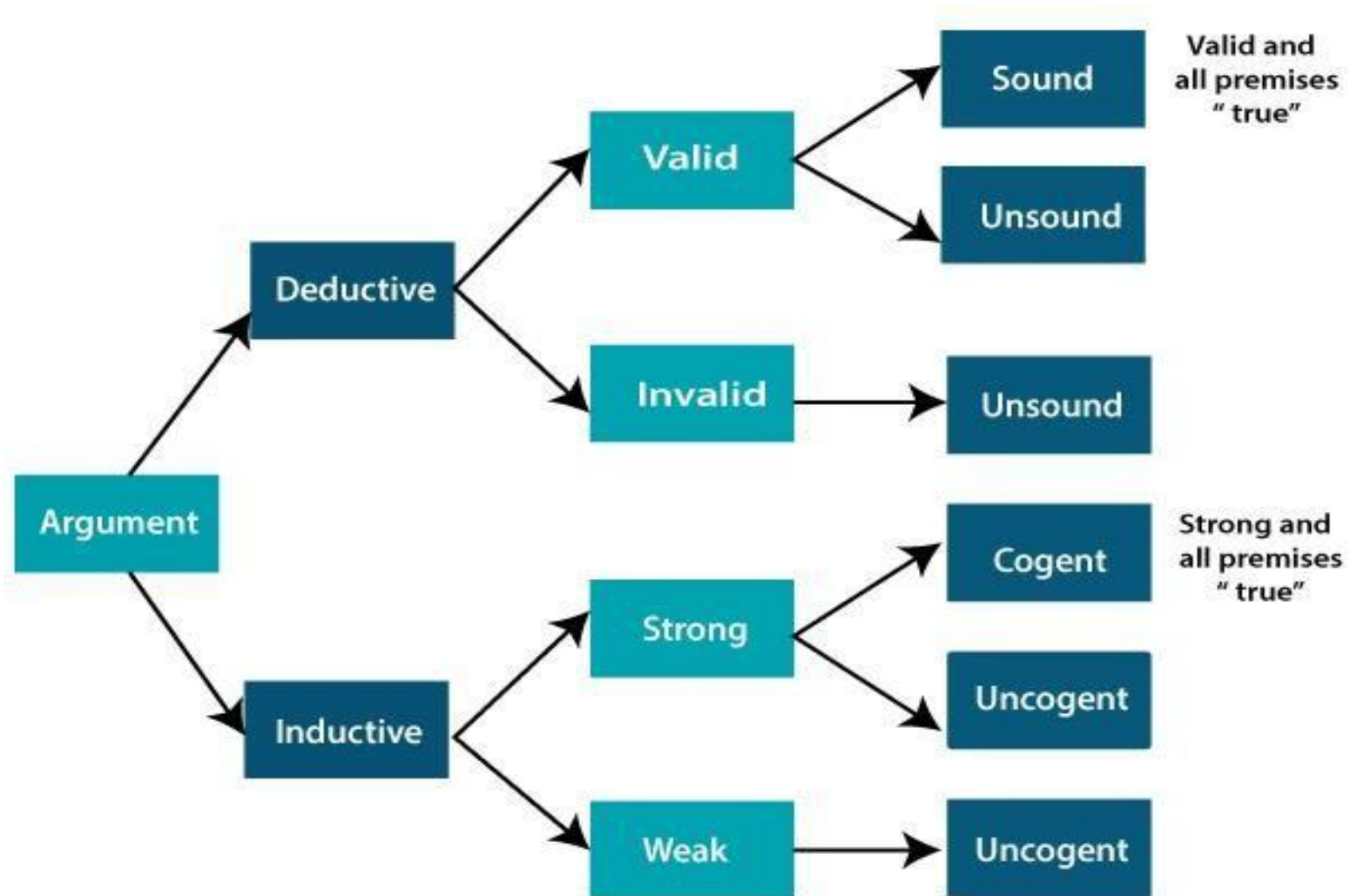
Difference between Inductive and Deductive reasoning

- Both reasoning forms have premises and conclusions, but both reasoning are contradictory to each other.
- Deductive reasoning uses available facts, information, or knowledge to deduce a valid conclusion,
 - whereas inductive reasoning involves making a generalization from specific facts, and observations.

...Cont.

- Deductive reasoning uses a **top-down approach**, whereas inductive reasoning uses a **bottom-up approach**.
- Deductive reasoning moves from generalized statement to a valid conclusion, whereas Inductive reasoning moves from specific observation to a generalization.
- In deductive reasoning, the **conclusions are certain**, whereas, in **Inductive reasoning, the conclusions are probabilistic**.
- Deductive arguments can be valid or invalid, which means if premises are true, the conclusion must be true, whereas inductive argument can be strong or weak, which means conclusion may be false even if premises are true.
- The differences between inductive and deductive can be explained using the below diagram on the basis of arguments:

...cont.



...cont.

Basis for comparison	Deductive Reasoning	Inductive Reasoning
Definition	Deductive reasoning is the form of valid reasoning, to deduce new information or conclusion from known related facts and information.	Inductive reasoning arrives at a conclusion by the process of generalization using specific facts or data.
Approach	Deductive reasoning follows a top-down approach.	Inductive reasoning follows a bottom-up approach.
Starts from	Deductive reasoning starts from Premises.	Inductive reasoning starts from the Conclusion.
Validity	In deductive reasoning conclusion must be true if the premises are true.	In inductive reasoning, the truth of premises does not guarantee the truth of conclusions.
Usage	Use of deductive reasoning is difficult, as we need facts which must be true.	Use of inductive reasoning is fast and easy, as we need evidence instead of true facts. We often use it in our daily life.
Process	Theory→ hypothesis→ patterns→confirmation.	Observations→patterns→hypothesis→Theory.
Argument	In deductive reasoning, arguments may be valid or invalid.	In inductive reasoning, arguments may be weak or strong.
Structure	Deductive reasoning reaches from general facts to specific facts.	Inductive reasoning reaches from specific facts to general facts.

Probabilistic reasoning in AI

- **Uncertainty:**

- KR using first-order logic and propositional logic with certainty, which means we were sure about the predicates.
- With this knowledge representation, we might write $A \rightarrow B$, which means if A is true then B is true, but consider a situation where we are not sure about whether A is true or not then we cannot express this statement, this situation is called uncertainty.
- So to represent uncertain knowledge, where we are not sure about the predicates, **we need uncertain reasoning or probabilistic reasoning.**

Uncertainty in Reasoning

- The world is an uncertain places; often the knowledge is imperfect which causes uncertainty. Therefore, reasoning must be able to operate under uncertainty.
- **AI systems must have ability to reason under conditions of uncertainty.**
 - **Incompleteness knowledge** → compensate for lack of knowledge.
 - **Inconsistencies knowledge** → Resolve ambiguities and contradictions
 - **Changing knowledge** → update the knowledgebase over time.

Causes of uncertainty

- Following are some **leading causes of uncertainty to occur in the real world.**
 - Information occurred from unreliable sources.
 - Experimental Errors
 - Equipment fault
 - Temperature variation
 - Climate change
- **Probabilistic reasoning:**

Probabilistic reasoning is a way of KR where we **apply the concept of probability to indicate the uncertainty in knowledge.**

 - In probabilistic reasoning, we **combine probability theory with logic to handle the uncertainty.**

...cont.

- As probabilistic reasoning uses probability and related terms, so before understanding probabilistic reasoning, let's understand some common terms:
- **Probability:** Probability can be defined as a chance that an uncertain event will occur.
- It is the numerical measure of the likelihood that an event will occur. The value of probability always remains between 0 and 1 that represent ideal uncertainties.

...cont.

- We use probability in probabilistic reasoning because:
 - it provides a way to handle the uncertainty that is the result of someone's laziness and ignorance.
- In the real world, there are lots of scenarios, where the certainty of something is not confirmed, such as
 - "It will rain today," "behaviour of someone for some situations," "A match between two teams or two players."
- These are probable sentences for which we can assume that it will happen but not sure about it, so here we use probabilistic reasoning.

...cont.

- **Need of probabilistic reasoning in AI:**
 - When there are unpredictable outcomes.
 - When specifications or possibilities of predicates becomes too large to handle.
 - When an unknown error occurs during an experiment.
- In probabilistic reasoning, there are two ways to solve problems with uncertain knowledge:
 - **Bayes' rule**
 - **Bayesian Statistics**

...cont.

- $0 \leq P(A) \leq 1$, where $P(A)$ is the probability of an event A
- $P(A) = 0$, indicates total uncertainty in an event A
- $P(A) = 1$, indicates total certainty in an event A
- The probability of an uncertain event by using the below formula.

$$\text{Probability of occurrence} = \frac{\text{Number of desired outcomes}}{\text{Total number of outcomes}}$$

- $P(\neg A)$ = probability of a not happening event.
- $P(\neg A) + P(A) = 1$.
- **Event:** Each possible outcome of a variable is called an event.

...cont.

- **Sample space:** The collection of all possible events is called sample space.
- **Random variables:** Random variables are used to represent the events and objects in the real world.
- **Prior probability:** The prior probability of an event is probability computed before observing new information.
- **Posterior Probability:** The probability that is calculated after all evidence or information has taken into account. It is a combination of prior probability and new information.

Conditional probability

- Conditional probability is a **probability of occurring an event when another event has already happened**.
 - Let's suppose, we want to calculate the event A when event B has already occurred, "the probability of A under the conditions of B", it can be written as:

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$

Where $P(A \wedge B)$ = Joint probability of a and B

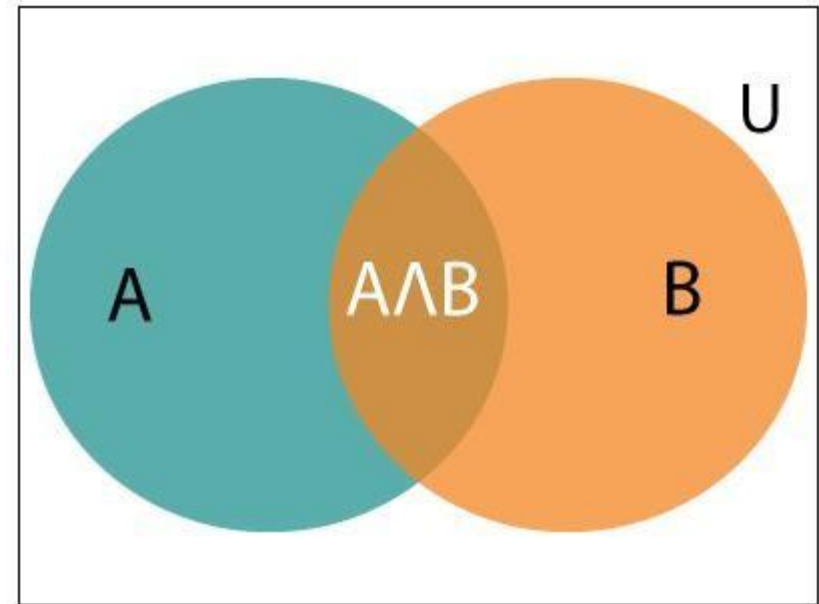
$P(B)$ = Marginal probability of B.

...cont.

- If the probability of A is given and we need to find the probability of B, then it will be given as:

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

- It can be explained by using the below Venn diagram, where B is occurred event, so sample space will be reduced to set B, and now we can only calculate event A when event B is already occurred by dividing the probability of **$P(A \cap B)$** by **$P(B)$** .



...cont.

■ **Example:**

- In a class, there are 70% of the students who like English and 40% of the students who likes English and mathematics, and then what is the percent of students those who like English also like mathematics?

■ **Solution:**

- Let, A is an event that a student likes Mathematics
- B is an event that a student likes English.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.4}{0.7} = 57\%$$

Hence, 57% are the students who like English also like Mathematics.

Thank You